

# Exposé X. Formalism of Intersections on Proper Algebraic Schemes

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(with an appendix by A. Grothendieck)

## 1. Compatibility of the filtrations with the composition law

$$K^\circ(X) \times K_\bullet(X) \longrightarrow K_\bullet(X).$$

### 1.1

First let  $X$  be a noetherian scheme, and let  $K_\bullet(X)$  be the group of classes of coherent  $\mathcal{O}_X$ -modules. For every coherent  $\mathcal{O}_X$ -module  $F$ , we denote its class in  $K_\bullet(X)$  by  $\text{cl}_\bullet(F)$ .

**Definition 1.1.1.** For every integer  $n \in \mathbb{Z}$ , let  $\text{Filt}_n(X)$  denote the subgroup of  $K_\bullet(X)$  generated by the classes  $\text{cl}_\bullet(F)$  of coherent  $\mathcal{O}_X$ -modules  $F$  such that

$$\dim(\text{Supp}(F)) \leq n.$$

This defines an increasing filtration  $(\text{Filt}_n(X))_{n \in \mathbb{Z}}$  of  $K_\bullet(X)$  such that  $\text{Filt}_n(X) = 0$  for  $n < 0$ .

**Proposition 1.1.2.** Let  $X$  be a noetherian scheme, let  $F$  be a coherent  $\mathcal{O}_X$ -module such that  $\dim(\text{Supp}(F)) \leq n$ , and, for  $n \in \mathbb{Z}$ , let  $X(n)$  be the subset

$$\{x \in X ; \dim(\overline{\{x\}}) = n\}$$

of  $X$ . For every  $x \in X$ , let  $Z(x)$  denote the integral closed subscheme associated with  $\overline{\{x\}}$ . Then the length of the  $\mathcal{O}_{X,x}$ -module  $F_x$  is finite for every  $x \in X(n)$  and is nonzero for only finitely many  $x \in X(n)$ , and one has

$$\text{cl}_\bullet(F) = \sum_{x \in X(n)} \text{length}(F_x) \text{cl}_\bullet(\mathcal{O}_{Z(x)}) + w, \quad (1.1.3)$$

where  $w \in \text{Filt}_{n-1}(X)$ .

By immediate induction on  $n$ , one obtains:

**Corollary 1.1.4.** For every noetherian scheme  $X$  and every integer  $n$ ,  $\text{Filt}_n(X)$  is the subgroup of  $K_\bullet(X)$  generated by the classes  $\text{cl}_\bullet(\mathcal{O}_Z)$ , where  $Z$  ranges over the integral closed subschemes of  $X$  of dimension at most  $n$ .

*Proof of 1.1.2.* Let  $X' = \text{Supp}(F)$ . Since  $X'$  is a noetherian space of dimension at most  $n$ , the elements of  $X' \cap X(n)$  are maximal in  $X'$  and hence finite in number; moreover,  $\text{length}(F_x)$  is finite for every  $x \in X' \cap X(n)$ . This proves the first assertion.

Let  $\mathcal{K} = \text{Coh}(X)$  be the category of coherent  $\mathcal{O}_X$ -modules, and let  $\mathcal{K}'$  be the subset of  $\text{Ob}(\mathcal{K})$  formed by the coherent  $\mathcal{O}_X$ -modules concentrated on  $X'$  that satisfy condition (1.1.3). Plainly  $\mathcal{K}'$  contains the sheaves  $\mathcal{O}_{Z(x)}$  for  $x \in X'$ . It is therefore enough to show that  $\mathcal{K}'$  is an “exact” subset of  $\text{Ob}(\mathcal{K})$ , for then the devissage lemma (EGA III 3.1.2) gives  $F \in \mathcal{K}'$ .

Thus let

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0 \quad (1.1.5)$$

be an exact sequence of  $\mathcal{O}_X$ -modules having two of its terms in  $\mathcal{K}'$ . The sheaves  $G'$ ,  $G$ , and  $G''$  are then all concentrated on  $X'$ . Hence the lengths  $\ell'_x$ ,  $\ell_x$ , and  $\ell''_x$  of the  $\mathcal{O}_{X,x}$ -modules  $G'_x$ ,  $G_x$ , and  $G''_x$  are finite for every  $x \in X(n)$ , and

$$\ell_x = \ell'_x + \ell''_x \quad (x \in X(n)). \quad (1.1.6)$$

Assertion (1.1.3) for all the terms of (1.1.5) now follows immediately from the hypothesis and the equation

$$\mathrm{cl}_\bullet(G) = \mathrm{cl}_\bullet(G') + \mathrm{cl}_\bullet(G''). \quad (1.1.7)$$

It is known that  $K_\bullet$  is a covariant functor on the category of locally noetherian schemes with respect to proper morphisms. The following proposition asserts that our filtration is functorial.

**Proposition 1.1.8.** Let  $f : X' \rightarrow X$  be a proper morphism of locally noetherian schemes, and let

$$f_* : K_\bullet(X') \longrightarrow K_\bullet(X)$$

be the morphism induced by  $f$ . Then

$$f_*(\mathrm{Filt}_n(X')) \subset \mathrm{Filt}_n(X) \quad (1.1.9)$$

for every integer  $n$ .

*Proof.* If  $F$  is a coherent  $\mathcal{O}_{X'}$ -module, then by definition (IV 2.11),

$$f_*(\mathrm{cl}_\bullet(F)) = \sum_{i \in \mathbb{N}} (-1)^i \mathrm{cl}_\bullet(R^i f_*(F)).$$

The  $\mathcal{O}_X$ -modules  $R^i f_*(F)$  are concentrated on  $f(\mathrm{Supp}(F))$ , and by EGA IV 5.4.1(ii),

$$\dim(\mathrm{Supp}(F)) \geq \dim(f(\mathrm{Supp}(F))),$$

which proves the assertion.

## 1.2

Let now  $X$  be an arbitrary scheme, and consider the group  $K^\circ(X)$  of classes of locally free  $\mathcal{O}_X$ -modules (henceforth understood to be of finite type). It is an augmented pre- $\lambda$ -algebra over the ring  $H^0(X, \mathbb{Z})$  (V 3.9.1). The map  $f \rightsquigarrow f \cdot 1 \in K^\circ(X)$  allows us henceforth to identify  $H^0(X, \mathbb{Z})$  with a subring of  $K^\circ(X)$ . If  $E$  is a locally free  $\mathcal{O}_X$ -module, we denote its class in  $K^\circ(X)$  by  $\mathrm{cl}^\circ(E)$ . The operators  $\lambda^k$  ( $k \in \mathbb{N}$ ) are defined by

$$\lambda^k(\mathrm{cl}^\circ(E)) = \mathrm{cl}^\circ(\wedge_{\mathcal{O}}^k E) \quad (1.2.1)$$

and by

$$\lambda^k(x + y) = \sum_{\substack{i+j=k \\ i,j \in \mathbb{N}}} \lambda^i(x) \lambda^j(y). \quad (1.2.2)$$

(V 2.2). Recall that in the  $\lambda$ -filtration  $(\mathrm{Filt}^i(X))_{i \in \mathbb{Z}}$  of  $K^\circ(X)$ , one has  $\mathrm{Filt}^i(X) = K^\circ(X)$  for  $i \leq 0$ , while for  $i \geq 1$ ,  $\mathrm{Filt}^i(X)$  is the ideal generated by the products

$$P = \gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k) \quad (1.2.3)$$

with  $1 \leq i_1, \dots, i_k$ ;  $i_1 + \cdots + i_k \geq i$ ; and  $x_1, \dots, x_k \in I = \mathrm{Ker}(\varepsilon)$ , where  $\varepsilon$  is the augmentation homomorphism (cf. Exposé V 3.10). It was proved in V 3.10 that  $\mathrm{Filt}^i(X)$  is even the  $H^0(X, \mathbb{Z})$ -submodule, hence, when  $X$  is connected, the abelian subgroup generated by the products (1.2.3), where one may moreover suppose that the elements  $x_p$  ( $p = 1, \dots, k$ ) have the form

$$x_p = \mathrm{cl}^\circ(E_p) - n_p, \quad (1.2.4)$$

with  $E_p$  a locally free  $\mathcal{O}_X$ -module of rank  $n_p \in H^0(X, \mathbb{Z})$ .

### 1.3

Recall (IV 2.10) that the bifunctor

$$(E, F) \rightsquigarrow E \otimes_{\mathcal{O}_X} F,$$

where  $E$  is a locally free  $\mathcal{O}_X$ -module and  $F$  is a coherent  $\mathcal{O}_X$ -module, induces a composition law

$$K^\circ(X) \times K_\bullet(X) \longrightarrow K_\bullet(X) \quad (1.3.1)$$

that makes  $K_\bullet(X)$  a unital  $K^\circ(X)$ -module. The key result of the present exposé is the following.

**Theorem 1.3.2.** Let  $X$  be a noetherian scheme. For all  $i, j \in \mathbb{Z}$ , the composition law (1.3.1) induces by restriction a composition law

$$\text{Filt}^i(X) \times \text{Filt}_j(X) \longrightarrow \text{Filt}_{j-i}(X). \quad (1.3.3)$$

**Corollary 1.3.3.** If  $X$  is a noetherian scheme of dimension at most  $j$ , then the canonical homomorphism

$$\Theta_X : K^\circ(X) \longrightarrow K_\bullet(X),$$

defined by  $\Theta_X(x) = x \cdot \text{cl}_\bullet(\mathcal{O}_X)$  ( $x \in K^\circ(X)$ ), induces for every  $i \in \mathbb{Z}$  a homomorphism

$$\text{Filt}^i(X) \longrightarrow \text{Filt}_{j-i}(X).$$

*Proof.* First let us show that (1.3.3), for  $i + j \leq k \in \mathbb{Z}$  and every integral closed subscheme of dimension at most  $j$ , implies Theorem 1.3.2 for  $i + j \leq k$ . By (1.1.4), it is enough to show that, for every  $x \in \text{Filt}^i(X)$  and every integral closed subscheme  $Z$  of dimension at most  $j$ , one has

$$x \text{cl}_\bullet(\mathcal{O}_Z) \in \text{Filt}_{j-i}(X).$$

To avoid confusion, let  $1_Z$  denote the class of  $\mathcal{O}_Z$  in  $K_\bullet(Z)$ . If  $\alpha : Z \rightarrow X$  is the canonical immersion, the projection formula (IV ...) gives

$$x \text{cl}_\bullet(\mathcal{O}_Z) = x \alpha_*(1_Z) = \alpha_*(\alpha^*(x) 1_Z).$$

Since  $\alpha^*(x) \in \text{Filt}^i(Z)$ , (1.1.8) and (1.3.3), applied with  $X = Z$  and  $i + j \leq k$ , give

$$\alpha_*(\alpha^*(x) 1_Z) \in \text{Filt}_{j-i}(X),$$

which proves the assertion.

Theorem 1.3.2 is trivial for  $i + j \leq 0$ . Suppose by induction that it has been proved for  $i + j < k$  and for every noetherian scheme  $X$ . By the preceding remark, it remains to prove that, for every integral noetherian scheme  $X$  of dimension  $j$  and every product (1.2.3) satisfying (1.2.4), one has

$$P \text{cl}_\bullet(\mathcal{O}_X) \in \text{Filt}_{j-i}(X). \quad (1.3.4)$$

If  $k > 1$  in (1.2.3), the assertion follows from the induction hypothesis and the associativity of the law (1.3.1). We may therefore suppose, since  $X$  is connected, that

$$x = P = \gamma^m(\text{cl}^\circ(E) - n), \quad (1.3.5)$$

where  $m \geq i$  and  $E$  is a locally free  $\mathcal{O}_X$ -module of rank  $n > 0$ . We distinguish two cases.

I. Suppose that there exist invertible  $\mathcal{O}_X$ -modules  $L_1, \dots, L_n$  such that

$$\text{cl}^\circ(E) = \text{cl}^\circ(L_1) + \dots + \text{cl}^\circ(L_n) \in K^\circ(X).$$